



Eco-Sync v2.0

VERSION	AUTHOR	DATE	STATUS
v2.0 (Triad Scope)	Dahunsi Samuel (DIVO)	May 2026	Architecture Locked

1. System Architecture: The Triad

Eco-Sync has evolved from a simple reporting app into a full-scale **Enterprise Resource Planning (ERP)** platform. The ecosystem is divided into three distinct operational layers that interact in real-time:

1. The User	2. The Admin	3. The Staff
The paying customer generating demand. Focuses on billing, bin requests, and triggering pickup alerts.	The strategic command. Focuses on HR, route generation, fleet management, and financial oversight.	The execution layer on the ground. Divided into Team Leads (Nav/Verification) and Normal Staff (Rosters).

2. User Personas & Feature Modules

2.1 THE USER (CUSTOMER HUB) CLIENT APP

Module	Functional Requirements
Onboarding & Provisioning	User registers full address (Street, House No.). Applies for route assignment. System generates an application ticket. Upon Admin approval, a unique Bin Number is provisioned and physical delivery is tracked in-app.

In-app wallet/payment integration for setup fees and monthly subscription renewals.

Module	Functional Requirements
Financial Gateway	
Service Triggers	One-tap "Bin Full" alarm. Receives proximity push notifications ("Team Titan is 2 blocks away").

2.2 THE ADMIN (COMMAND CENTER) WEB DASHBOARD

Module	Functional Requirements
Dynamic Dispatch	Receives critical alerts when a route hits "Critical Mass" of full bins. Assigns available teams to the route.
HR & Fleet Ops	Create/manage "Teams". Add/remove staff. Assign/demote Team Leads. Build weekly duty rosters and shifts.
Application CRM	Review new user signups, verify addresses, approve applications, and trigger bin delivery logistics.

2.3 THE STAFF (GROUND OPS) FIELD APP

Team Leads (Supervisors)

- **Route Navigation:** Receives critical dispatch alerts. Opens in-app GPS routing mapping the fastest path through the assigned route.
- **Resolution Protocol:** Marks bins as resolved. *(See Section 3 for verification protocol).*

Normal Staff (Field Workers)

- **Duty Roster:** Views assigned shifts, dates, and designated Team Lead.
- **Attendance Logic:** GPS-gated clock-in/clock-out functionality at the depot.

3. Advanced Logic & Enterprise Protocols

To ensure business viability and operational truth, Eco-Sync relies on three advanced logic gates:

A. The Route-Health Algorithm (Auto-Scaling)

Admins do not manually hunt for full bins. The system groups houses by `route_id`. Once 70% of users on "Route Alpha" trigger the "Bin Full" alarm, the system automatically elevates the route to **CRITICAL** status and recommends dispatch.

B. QR/NFC Proof of Service (Anti-Fraud)

Staff cannot blindly click "Resolved" from the truck. Every physical bin contains a registered QR code. The Team Lead must scan the QR code to clear the bin from the map, triggering an automatic SMS/Push to the User confirming the exact pickup time.

C. Defaulter Logic (Revenue Protection)

If a User fails to pay their monthly fee:

1. Their app access is restricted to the Payment Gateway.

- 2. The "Bin Full" alarm is hard-disabled.
- 3. On the Team Lead's navigation map, the defaulter's house is rendered in **RED** with a "SKIP: UNPAID" tag.

4. Expanded Data Schema (Supabase)

Table	Key Fields	Purpose
users	uid, role, status (active/defaulter), route_id	Identity & billing status.
bins	bin_id, user_id, qr_hash, fill_status	Tracks physical hardware.
teams	team_id, lead_id, truck_id, current_route	HR team grouping.
shifts	shift_id, staff_id, date, clock_in_time, status	Payroll and attendance.
routes	route_id, name, polyline, current_capacity	The Route-Health engine.